

# Mileage Tracker v2 — Build Report

**Story:** story\_42d9ba — Build Mileage Tracker v2 from requirements with design system  
**Epic:** epic\_4eb084 — Rebuild Mileage Tracker (v2)  
**Repository:** raghu-cmyk/mileage-tracker-v2  
**Branch:** feat/mvp-v2 → mc\_webhook\_test (PR #8)  
**Generated:** 20260608T194042Z

## 1. Job Summary

| Order | Job ID                  | Type      | State     | Duration    |
|-------|-------------------------|-----------|-----------|-------------|
| 1     | 20260608T192719Z_c87162 | plan      | completed | 38s         |
| 2     | 20260608T192923Z_590968 | implement | completed | 2m 51s      |
| 3     | 20260608T193424Z_5031ca | validate  | completed | 1m 1s       |
| 4     | 20260608T193903Z_0b40a5 | deliver   | completed | 25s         |
| 5     | 20260608T193952Z_6ab95f | report    | running   | in progress |

## 2. Requirements → GitHub Issues Mapping

| GitHub Issue                                       | Requirements Covered                                         |
|----------------------------------------------------|--------------------------------------------------------------|
| #1 Authentication & Secure Session Management      | FR-A1–FR-A4, NFR-3, NFR-4 (§5.1, §6.2)                       |
| #2 Vehicle Management & Annual Odometer Readings   | §4.2, FR-V1–FR-V2, AuditEvent (§5.2)                         |
| #3 Trip Management with §274(d) Substantiation     | §4.3, FR-T1–FR-T5, FR-C1–FR-C2 (§5.3–5.4)                    |
| #4 Trip Categories & Time-Effective IRS Rates      | §4.4, §4.5, NFR-1, NFR-2, NFR-5                              |
| #5 Deduction Calculation & Year-End Summary        | FR-R1–FR-R3, NFR-2, NFR-5 (§5.5)                             |
| #6 IRS-Compliant CSV & PDF Export                  | FR-E1–FR-E3, NFR-7 (§5.6)                                    |
| #7 Design & Aesthetic — Ledger Slate Design System | Visual design system (color, typography, layout, components) |

### 3. Per-Job Agent Activity Logs

#### Job: 20260608T192719Z\_c87162 (plan)

```
## Planning Complete
Read `docs/requirements/mileage-tracker-requirements.md` from **raghu-cmyk/mileage-tracker-v2** and decomposed the
MVP into **7 GitHub issues** (6 feature tasks + 1 design system).
### Feature Issues (`plan-task`)
| # | Issue | URL |
|---|---|---|
| 1 | **Authentication & Secure Session Management** | https://github.com/raghu-cmyk/mileage-tracker-v2/issues/1 |
| 2 | **Vehicle Management & Annual Odometer Readings** | https://github.com/raghu-cmyk/mileage-tracker-v2/issues/2 |
| 3 | **Trip Management with §274(d) Substantiation & Audit Trail** | https://github.com/raghu-cmyk/mileage-tracker-v2/issues/3 |
| 4 | **Trip Categories & Time-Effective IRS Mileage Rates (Reference Data)** | https://github.com/raghu-cmyk/mileage-tracker-v2/issues/4 |
| 5 | **Deduction Calculation & Year-End Summary** | https://github.com/raghu-cmyk/mileage-tracker-v2/issues/5 |
| 6 | **IRS-Compliant CSV & PDF Export** | https://github.com/raghu-cmyk/mileage-tracker-v2/issues/6 |
### Design Issue (`design`)
| # | Issue | URL |
|---|---|---|
| 7 | **Design & Aesthetic – Mileage Tracker v2 Visual Design System** ("Ledger Slate" palette) | https://github.com/raghu-cmyk/mileage-tracker-v2/issues/7 |
### Decomposition Summary
The backlog maps directly to the requirements spec:
- **Auth** (§5.1) – single-user, Argon2id, sessions, rate limiting
- **Vehicles + odometer** (§4.2, §5.2) – CRUD, archive-only delete, tax-year odometer readings
- **Trips** (§4.3, §5.3–5.4) – full §274(d) fields, late-entry flagging, audit trail
- **Reference data** (§4.4–4.5) – categories + time-effective IRS rates (2025/2026), DB-seeded not hardcoded
- **Reporting** (§5.5) – deduction math, business-use %, year-end summary
- **Export** (§5.6) – CSV trip log + PDF summary for accountant
- **Design system** – color tokens, Inter/JetBrains Mono typography, layout grid, component specs, accessibility
All issues reference branch `feat/mvp-v2` (from `mc_webhook_test`).
...

mc done --summary "Created 7 planning issues (incl. design) for Build Mileage Tracker v2 from requirements with
design system"
...
```

#### Job: 20260608T192923Z\_590968 (implement)

```
Implemented **story_42d9ba** – the full Mileage Tracker v2 MVP.
## Delivered
**Target:** `raghu-cmyk/mileage-tracker-v2` on branch `feat/mvp-v2`
| Capability | Source |
|---|---|
| Authentication (Argon2id, sessions, rate limiting) | Issue #1 |
| Vehicle management + odometer readings | Issue #2 |
| Trip CRUD with §274(d) substantiation + audit trail | Issue #3 |
| Time-effective IRS rates (DB reference data) | Issue #4 |
| Deduction calculation + year-end summary | Issue #5 |
| CSV + PDF export | Issue #6 |
| Ledger Slate design system | Issue #7 |
## Verification
- **19 tests passed** (`pytest -q`)
- App imports cleanly: `Mileage Tracker v2` 2.0.0
- **No TODO/FIXME markers** in workspace source (acceptance gate)
- Build report: `docs/build-reports/mvp-v2-build-report.pdf`
## Artifacts
- **PR:** https://github.com/raghu-cmyk/mileage-tracker-v2/pull/8 (`feat/mvp-v2` → `mc_webhook_test`)
- **Commit:** `6a4a19d` on `feat/mvp-v2`
- **Output:** `artifacts/output.json` in job workspace
```

- **mc report/done** called for job `20260608T192923Z\_590968`  
The implementation adapts the proven v1 FastAPI stack and applies the Ledger Slate design tokens (Inter + JetBrains Mono, WCAG-aligned palette, responsive layout) in `app/templates/base.html`.

## Job: 20260608T193424Z\_5031ca (validate)

**Validation passed** for `feat/mvp-v2` on `raghu-cmyk/mileage-tracker-v2`.

## Summary

The candidate branch was cloned, dependencies installed, and the app validated in-process using FastAPI's `TestClient` (no long-lived server).

### Results

| Check                        | Result                             |
|------------------------------|------------------------------------|
| App import                   | `Mileage Tracker v2` loads cleanly |
| In-process route checks      | <b>12/12 passed</b>                |
| Unit tests (`pytest tests/`) | <b>19/19 passed</b>                |

### Routes exercised

- `/health` → 200
- `/` → 303 redirect to `/login` (unauthenticated)
- `/login`, `/register` → 200
- POST /register → 303 redirect to dashboard
- `/` (authenticated), `/vehicles`, `/trips`, `/summary` → 200
- `/api/summary/2026` → 200
- `/vehicles/new`, `/trips/new` → 200

### Artifacts

- `validation\_evidence.json` – route status codes and results
- `validation\_tests.log` – full pytest output
- VALIDATION\_PASSED – one-line pass summary
- mileage-tracker-v2/ – cloned candidate branch

Mission control job marked complete: `mc done --job 20260608T193424Z\_5031ca`.

## Job: 20260608T193903Z\_0b40a5 (deliver)

## Delivery complete

Validated branch and pull request are ready for review.

### Branch confirmation

| Check                           | Status                                                 |
|---------------------------------|--------------------------------------------------------|
| Branch `feat/mvp-v2` exists     | Yes                                                    |
| Branched from `mc_webhook_test` | Yes (parent: `e2f2c899`)                               |
| MVP commit                      | `6a4a19d` – full implementation + tests + build report |
| Build report                    | `docs/build-reports/mvp-v2-build-report.pdf` on branch |

### Pull request

**URL:** <https://github.com/raghu-cmyk/mileage-tracker-v2/pull/8>

| Field        | Value                                       |
|--------------|---------------------------------------------|
| <b>Head</b>  | `feat/mvp-v2`                               |
| <b>Base</b>  | `mc_webhook_test` (not `main`)              |
| <b>Title</b> | Mileage Tracker v2 – Full MVP (feat/mvp-v2) |
| <b>State</b> | Open, ready for review                      |

The PR body was updated with validation details, follow-ups, and `Closes #1` through `Closes #7` for the planning issues (auth, vehicles, trips, rates, deductions, exports, design system).

### What was delivered

- Authentication (Argon2id, sessions, rate-limited login)
- Vehicles + per-tax-year odometer readings
- Trips with IRC §274(d) substantiation and audit trail
- Time-effective IRS mileage rates as DB reference data (2025/2026)
- Deduction calculation and year-end summary
- IRS-compliant CSV and PDF export
- Ledger Slate design system across templates
- 19 passing unit tests; app starts without errors

### Mission Control

```

```
mc done --summary "Opened PR for Build Mileage Tracker v2 from requirements with design system into  
mc_webhook_test"
```

```

## 4. Validation Results (Gate Outcomes)

| Stage     | Job ID                  | Gate                            | Outcome                          |
|-----------|-------------------------|---------------------------------|----------------------------------|
| plan      | 20260608T192719Z_c87162 | gate_min_issues.sh (>=3 issues) | PASS — 7 issues created          |
| implement | 20260608T192923Z_590968 | no_todo_markers (workspace)     | PASS — no placeholder markers    |
| validate  | 20260608T193424Z_5031ca | VALIDATION_PASSED file          | PASS — 12/12 routes, 19/19 tests |
| deliver   | 20260608T193903Z_0b40a5 | gate_pr_open.sh mc_webhook_test | PASS — PR #8 open                |
| report    | 20260608T193952Z_6ab95f | report-*.pdf (>2k)              | PENDING                          |

### Validate-stage evidence (validation\_evidence.json):

```
/health: HTTP 200 [OK] — body={'status': 'ok'}
/: HTTP 303 [OK] — location=/login
/login: HTTP 200 [OK] — page renders
/register: HTTP 200 [OK] — page renders
POST /register: HTTP 303 [OK] — location=/
/ (authenticated): HTTP 200 [OK] — dashboard renders
/vehicles: HTTP 200 [OK] — page renders
/trips: HTTP 200 [OK] — page renders
/summary: HTTP 200 [OK] — page renders
/api/summary/2026: HTTP 200 [OK] — tax_year=2026
/vehicles/new: HTTP 200 [OK] — form renders
/trips/new: HTTP 200 [OK] — form renders
```

VALIDATION\_PASSED: Validated feat/mvp-v2: app imports cleanly, 12/12 routes return expected status codes via in-process TestClient, 19/19 unit tests pass.

## 5. Total Build Time

**Chain start:** 2026-06-08T19:27:19Z (plan job claimed)

**Chain end (pre-report):** 2026-06-08T19:39:29Z (deliver job completed)

**Elapsed:** 12 minutes 10 seconds

**Per-stage durations:** plan 38s · implement 2m 51s · validate 1m 1s · deliver 25s

## 6. Application Code Statistics

| Metric                    | Value                                                                                                                         |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Repository                | raghu-cmyk/mileage-tracker-v2                                                                                                 |
| Branch                    | feat/mvp-v2                                                                                                                   |
| Pull Request              | <a href="https://github.com/raghu-cmyk/mileage-tracker-v2/pull/8">https://github.com/raghu-cmyk/mileage-tracker-v2/pull/8</a> |
| Total files (excl. .venv) | 45                                                                                                                            |
| Total lines of code       | 3742                                                                                                                          |
| Python lines of code      | 2458                                                                                                                          |
| Unit tests passing        | 19/19                                                                                                                         |